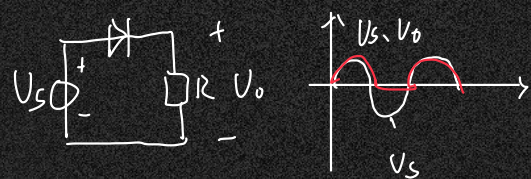


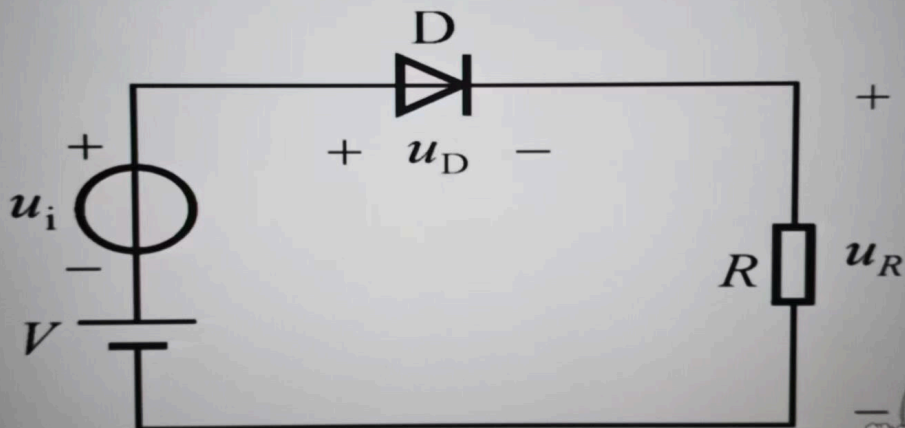
$U_S - U_1 \leq U_{on}$ 即 $U_S \leq U_1 + U_{on}$ 时截止

$\Rightarrow U_O = U_S$

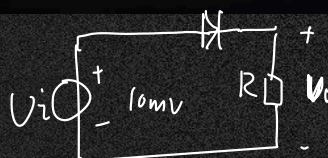
当 $U_S > U_1 + U_{on}$, $U_O = U_1 + U_{on}$



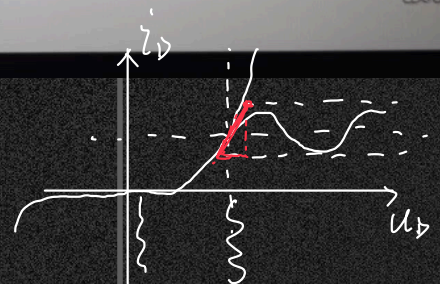
直流电压源和交流电压源同时作用的二极管电路



整流



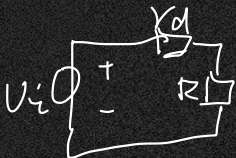
$$i_D = I_S (e^{\frac{U_D}{U_T}} - 1)$$



$$\frac{\Delta U_D}{\Delta i_D} = r_d$$

$$I_D = \frac{U - U_{on}}{R}$$

$$r_d = \frac{U_T}{I_D} = \frac{U_T \cdot R}{U - U_{on}}$$



$$i_d = \frac{U_i}{R_d + R}$$

$$i_D = i_d + I_D$$

(直流都有 = 交 + 直)

$$i_d = \frac{U_i}{R_d + R}$$

$$\frac{1}{r_d} = \frac{\Delta i_D}{\Delta U_D} = \frac{d i_D}{d U_D} = I_S \cdot \frac{1}{U_T} e^{\frac{U_D}{U_T}}$$

$$= \frac{I_S e^{\frac{U_D}{U_T}}}{U_T} \approx \frac{I_D}{U_T}$$

$$\Rightarrow r_d = \frac{U_T}{I_D}$$

三、二极管的微变等效

1. 直流通 $\rightarrow I_D$

$$2. r_d = \frac{U_T}{I_D}$$



二极管的微变等效电路图

